dream and beyond—a theory, I and many others believe, that is blazing the beginnings of a trail which will one day lead us to the deepest laws of the universe. Truthfully, though, superstring theory was not conceived as an ingenious means to reach these noble and long-standing goals. Instead, the history of superstring theory is full of accidental discoveries, false starts, missed opportunities, and nearly ruined careers. It is also, in a precise sense, the story of the discovery of the right solution for the wrong problem.

In 1968, Gabriele Veneziano, a young postdoctoral research fellow working at CERN, was one of many physicists trying to understand the strong nuclear force by studying the results of high-energy particle collisions produced in atom smashers around the world. After months of analyzing patterns and regularities in the data, Veneziano recognized a surprising and unexpected connection to an esoteric area of mathematics. He realized that a two-hundred-year-old formula discovered by the famous Swiss mathematician Leonhard Euler (the *Euler beta function*) seemed to match data on the strong nuclear force with precision. While this might not sound particularly unusual—theoretical physicists deal with arcane formulae all the time—it was a striking case of the cart's rolling miles ahead of the horse. More often than not, physicists first develop an intuition, a mental picture, a broad understanding of the physical principles underlying whatever they are studying and only then seek the equations necessary to ground their intuition in rigorous mathematics. Veneziano, to the contrary, jumped right to the equation; his brilliance was to recognize unusual patterns in the data and to make the unanticipated link to a formula devised centuries earlier for purely mathematical interest.

But although Veneziano had the formula in hand, he had no explanation for *why* it worked. He lacked a physical picture of why Euler's beta function should be relevant to particles influencing each other through the strong nuclear force. Within two years the situation completely changed. In 1970, papers by Leonard Susskind of Stanford, Holger Nielsen of the Niels Bohr Institute, and Yoichiro Nambu of the University of Chicago revealed the physical underpinnings of Veneziano's discovery. These physicists showed that if the strong force between two particles were due to a tiny, extremely thin, almost rubber-band-like strand that connected the particles, then the quantum processes